

When Does Action-Type Supervision Help GUI Grounding? Separating Real Effects from Evaluation Artifacts

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Abstract

GUI agents must predict an action type (click, scroll, type, ...) and a spatial target from a screenshot and an instruction. We study whether explicitly factoring these decisions helps a 2B-parameter agent, via a matched-compute comparison of five conditioning mechanisms on Qwen2-VL: a flat baseline (A), an auxiliary action-type loss (B), hard routing (C), an additive embedding hook (D-hook), and the prepended learned action embedding our original proposal advocated (D-token). Our first evaluation of this question was wrong in three independent ways, each of which we diagnose, quantify, and repair: (i) injecting conditioning via `inputs_embeds` silently bypasses Qwen2-VL’s M-RoPE position path and manufactured an 8σ false negative; (ii) train/validation splits taken as consecutive slices of one episode stream leaked shared task goals (24% of validation steps at the headline setting, up to 85% at scale) while looking clean at the episode level (2%); and (iii) a paired bootstrap pooling (seed, example) predictions as independent units was anti-conservative, rejecting a true clustered null 21.2% of the time at nominal 5% (an episode-clustered replacement achieves 5.1%). We then rebuilt the data pipeline around frozen goal-disjoint splits with a test region no phase of the project had ever touched, pre-specified four contrasts, and evaluated once. Every mechanistic conclusion of the original study reversed, and the statistical repair alone would not have caught most of them. The auxiliary loss that the repaired statistics still endorsed on historical data (B–A $+0.051$ [$+0.020$, $+0.081$], $p = 0.0043$, versus the invalid pooled $p = 0.0002$) is null on the untouched test set ($+0.011$ [-0.005 , $+0.029$], Holm $p = 0.309$); the prepended action embedding our proposal advocated and the course-era version reported as refuted in fact beats it (D-token–B $+0.035$ [$+0.017$, $+0.054$], Holm $p = 0.0027$) — though D-token reads the ground-truth action type at test time and B reads nothing, so that contrast is oracle-versus-deployable; the largest relative gain moves to scroll ($2.5\times$), while click still supplies 61% of the absolute gain; the control condition designed to show no effect

shows one, and is therefore invalid; and the deployable predicted-type pipeline is null ($+0.015$ [-0.004 , $+0.034$], Holm $p = 0.309$) because Stage 1 accuracy falls from 0.795 on the frozen validation split to 0.708 on the untouched test region, absorbing 76% of the oracle gain and leaving no detectable benefit. What survives is narrow and honest: oracle action-type conditioning improves grounding (D-hook–A $+0.057$ [$+0.035$, $+0.079$], Holm $p = \leq 0.0004$), and nothing we can actually deploy does. We release the frozen splits, per-example predictions, and the clustered-inference code. A study can be multi-seed, ablated, controlled, and apparently significant at the pooled $p=0.0002$ we now retract, and still have every one of its mechanistic claims move under a clean split and a calibrated test.

1. Introduction

A GUI agent maps a screenshot and a natural-language instruction to a low-level action: a type (click, scroll, type, drag, ...) and, for spatial actions, a pixel-level target. State-of-the-art native agents (UITARS [13], OS-Atlas [11], SeeClick [3], ShowUI [6]) emit both in one autoregressive stream, entangling the two decisions. A documented failure mode follows: commit to the wrong action type, then ground coherently within the wrong semantic frame – scrolling past a visible target instead of clicking it. This motivated our original hypothesis, formulated as an architecture bet: a learned action-type embedding, prepended to the instruction, should let a grounding model condition on what before deciding where, and should help most in low data.

This paper reports what happened when we tested that bet carefully, and it has two intertwined subjects. The first is the scientific question: when does action-type supervision change GUI grounding, and through which mechanism? The second is methodological: which implementation and evaluation choices can create a false appearance of benefit or harm in studies of exactly this kind? We came to the second subject involuntarily – three separate artifacts in our own pipeline each produced a wrong conclusion at some

point during the project – and we believe the post-mortem is at least as useful to practitioners as the object-level result.

Concretely, we compare five matched-compute Stage-2 conditioning mechanisms on Qwen2-VL-2B [10] with LoRA, over AITW [9] with a Mind2Web [4] cross-dataset control, preceded by a Stage-1 action-type classifier study. Our contributions:

- A confirmatory answer, and it is mostly negative. On an untouched, goal-disjoint test set: auxiliary action-type supervision does not measurably help; oracle inference-time conditioning does (+0.057 [+0.035, +0.079]); the deployable predicted-type pipeline does not, because the action-type classifier drops from 0.795 to 0.708 accuracy between the two held-out regions; and the proportionally largest effect moves to scroll rather than the click disambiguation our exploratory analysis reported, though click’s higher frequency still makes it 61% of the absolute gain.
- Three quantified evaluation pitfalls. (1) A silent M-RoPE / inputs_embeds interaction that fabricated an 8σ false negative; (2) episode-stream train/validation leakage that is invisible at the episode level (2% of validation steps) but large at the shared-task-goal level (24%–85%); (3) pseudo-replicated significance testing whose anti-conservativeness we measure on synthetic clustered nulls (21.2% type-I error at nominal 5%).
- A repaired, releasable protocol. Pinned-revision, goal-disjoint frozen train/validation/test manifests with an untouched test region beyond every historical data access; episode-clustered paired inference with per-seed effects and validated calibration; and a prespecified confirmatory analysis committed before test access. All artifacts, per-example predictions, and analysis code are released.
- A measured cost of getting evaluation wrong. Because we ran the same ablation twice — once under the flawed protocol, once under the repaired one — we can report what the flaws actually bought: not inflated p -values alone, but a reversal of which mechanism wins, which action class carries the gain, whether our own hypothesis was refuted, whether the pipeline is deployable, and whether the control holds (Section 7.4).

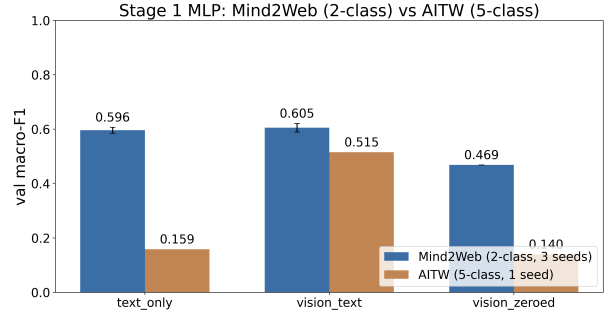


Figure 1. Stage 1 action-type classifier macro-F1 on Mind2Web (2 active classes) and AITW (5 active classes). Vision is essential on AITW (+0.414 over text-only) but redundant on Mind2Web (+0.009), because AITW instructions describe goals rather than physically specifying the action.

2. Related Work

End-to-end GUI agents. The dominant paradigm treats action type, arguments, and coordinates as one autoregressive output. UI-TARS [13] exemplifies this with a unified action space across platforms and explicit thought traces. OS-Atlas [11] scales the same paradigm with 13M cross-platform GUI grounding instances and shows that action-label harmonization across web/Android/desktop is nontrivial. ShowUI [6] builds a compact VLA around Qwen2-VL with interleaved vision-language-action streaming. Our flat baseline (Variant A) replicates the joint-decoding paradigm of these systems.

Modular grounding. SeeClick [3] argues that grounding is the central bottleneck in visual GUI agents. UGround / SeeAct-V [5] separates planning from grounding at the text-vs-pixel boundary. GUI-Actor [14] replaces coordinate-token regression with an attention-based action head driven by a dedicated <ACTOR> token plus a verifier. Our work studies a different module boundary: between action-type prediction and spatial grounding, not between planning text and coordinates.

Reward decomposition. UI-R1 [7] applies RL fine-tuning with a rule-based reward that separates action-type accuracy from action-argument accuracy on AndroidControl. It is the closest training analog: type and location as separable optimization targets; we use supervised conditioning rather than RL. Region-Focus [8] refines grounding at inference; we change the training-time conditioning pathway. SayCan [1] and ReAct [12] provide the modular-decision precedent, and RT-2 [2] the unified token-level action generation paradigm we critically examine.

Evaluation integrity. Our methodological findings echo, in a GUI setting, long-standing concerns about test-set contamination and dependence-blind inference: grouped data require grouped splits and cluster-aware uncertainty, and repeated measurements of the same example must not be pooled as independent evidence. Mind2Web’s own metric design (element accuracy vs. operation F1 vs. step success) [4] reflects the type/location decomposition we study.

3. Data and Splits

3.1. Sources and preprocessing

We use two datasets under a canonical 8-class action taxonomy (click, double_click, type, scroll, drag, hotkey, wait, finished).

AITW [9] (via the cjcfsjt/AITW_General Hugging-Face mirror, revision pinned to 5c0dc713) is the primary Stage-2 benchmark: five active classes (click 55%, scroll 13%, finished 12%, type 11%, hotkey 9%). Screenshots in the mirror are raw RGB bytes without headers; we decode them against seven observed resolutions with a portrait-aspect fallback. DUAL_POINT events are split into taps (click) versus swipes (scroll) by normalized touch-lift distance (threshold 0.04). Two data mixes matter below: all_with_coords (tap, swipe, type – action type is spatially predictive because taps target small elements while swipes are gestures) and taps_and_swipes (a control where knowing the type carries little information about where).

Mind2Web [4] (multimodal release) provides the Stage-1 cross-dataset comparison and a Stage-2 cross-dataset control; it is click/type-dominated with only two active classes at scale.

Coordinates. Targets are normalized to $[0, 1]^2$ and rendered as the string "(x, y)" on a $[0, 999]$ grid; training loss is masked to the coordinate tokens.

3.2. Two evaluation protocols

Exploratory protocol (Phases 1–7, historical). Our original runs streamed the AITW training split in mirror order, filtered by data mix, and took consecutive slices: the first n_{train} steps for training and the next n_{val} for validation. This is the kind of split that looks innocuous in code review and is not: episodes and, worse, repeated task goals straddle the boundary (Section 5.2). Every historical number in this paper is labeled exploratory and is re-analyzed with dependence-aware statistics.

Confirmatory protocol (Phase 9, frozen). We rebuilt the data pipeline around a frozen manifest: the first 12,000 all_with_coords steps of the pinned mirror revision, grouped by normalized task goal (episodes

never span goals), split goal-disjointly into train (9,511 steps), validation (1,660), and an untouched test region (829): test groups are drawn exclusively from beyond stream index 5,500, past the furthest index any historical run ever loaded (5,250), so no test step was trained on, evaluated on, or inspected during the exploratory phase. Evaluation sets are frozen and identical for every variant and training size (250 validation / 600 test steps for all_with_coords; 200/400 for the control mix); training subsets are nested prefixes. The manifest, split hashes, 17 disclosed near-duplicate goal pairs (token Jaccard ≥ 0.8), duplicate checks, and the prespecified analysis (contrasts, metrics, corrections, failed-run rules) were committed to version control before any test-set evaluation ran. We claim “decisions frozen before test access,” documented by commit history – not third-party preregistration.

4. Methods

4.1. Stage 1: action-type classifier

A frozen Qwen2-VL-2B encodes screenshot+instruction; mean-pooled final hidden states (1536-d) feed a 3-layer MLP trained with class-weighted cross-entropy (AdamW, lr 10^{-4} , batch 128, 8 epochs, best epoch by validation macro-F1). Modes: text_only vs. vision_text.

4.2. Stage 2: five conditioning mechanisms

Stage 2 fine-tunes Qwen2-VL-2B with LoRA (rank 16 on q/k/v/o projections, $\approx 0.2\%$ of parameters) to emit the coordinate string; all variants share data, optimizer (lr 2×10^{-5} , 2 epochs, batch 1), and evaluation. The only difference is how action-type information enters:

A (flat baseline). No conditioning; replicates joint decoding.

B (auxiliary loss). A linear action-type head on the pooled representation adds cross-entropy to the grounding loss ($\lambda=1$); no action information at inference.

C (hard routing). Trained to emit "{word} (x, y)"; at inference the gold action word is forced as the decode prefix.

D-hook (additive embedding). A learned table $E \in \mathbb{R}^{8 \times d}$; the row for the current action type is added to all token embeddings via a forward hook, preserving the normal input_ids path (embedding norm at convergence ≈ 0.07 at the headline setting).

D-token (prepended token – the original hypothesis). A sentinel $\langle \text{action_slot} \rangle$ token is prepended to the instruction and its embedding replaced in place by $E[a]$, M-RoPE-correctly (the token is in input_ids);

norm at convergence ≈ 2.2 , a strong routable signal.

Objective. With screenshot s , instruction g , action type a , coordinate tokens $y_{1:T}$ and variant-specific conditioning c :

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, s, g, c), \quad (1)$$

with $c = \emptyset$ for A and B, $c = E[a]$ for D-hook/D-token, and the forced word for C; B adds $\lambda \text{CE}(h_{\phi}(\bar{z}), a)$ (Eq. 1 otherwise unchanged).

End-to-end pipeline. The deployable configuration trains D-hook, then a small MLP head on frozen-backbone features (Stage 1 in-job), and evaluates grounding under predicted types; gold-type results are oracle ceilings, never deployable claims.

4.3. Evaluation metrics

hit@ r : fraction of predictions within normalized L2 distance r of the target ($r \in \{0.05, 0.10, 0.25\}$; headline 0.10). **Mean L2:** mean normalized distance (parse failures scored at the $\sqrt{2}$ sentinel and counted as misses; parse rates reported). Statistical methodology is Section 5.3.

5. Three Evaluation Pitfalls, Quantified

5.1. Pitfall 1: M-RoPE vs. inputs_embeds

Qwen2-VL assigns 3D (temporal, height, width) position IDs to visual tokens via an `input_ids`-keyed path. Our first D-token implementation injected the action embedding by passing `inputs_embeds` directly, which silently bypasses that path: image-token positions were computed without the action slot in the sequence, misaligning multimodal position embeddings for every visual token. The observed effect was a confident 8σ “A beats D” result – a pure artifact. Rebuilding the same mechanism through the `input_ids` path (D-hook, D-token) erased the deficit. Any conditioning study on M-RoPE-family models that touches `inputs_embeds` should verify position-ID computation explicitly; the artifact is silent, large, and directionally misleading.

5.2. Pitfall 2: stream-slice splits leak task goals

Table 1 quantifies the historical protocol’s leakage by reconstructing, for every historical cell, which episodes and normalized task goals each validation step shared with that cell’s training slice. Episode-level overlap – the leakage one usually checks – is small (2% at the headline cell). The dominant channel is shared task goals: 24% of headline validation steps have a

historical cell	val steps	sharing a train episode	sharing a train goal
all_wc, $n=300$	250	4 (2%)	14 (6%)
all_wc, $n=500$	250	4 (2%)	47 (19%)
all_wc, $n=800$	250	5 (2%)	49 (20%)
all_wc, $n=1200$	250	4 (2%)	61 (24%)
all_wc, $n=2500$	250	10 (4%)	133 (53%)
all_wc, $n=5000$	250	2 (1%)	213 (85%)
taps_swipes, $n=1000$	200	0 (0%)	48 (24%)

Table 1. Leakage in the historical (exploratory) splits: fraction of validation steps sharing an episode or a normalized task goal with that cell’s training slice. Goal sharing, not episode straddling, is the dominant channel.

goal string that also appears in training, rising to 85% at $n_{\text{train}}=5000$ – AITW goals repeat across many episodes, so consecutive-slice splitting almost guarantees template overlap at scale. Val-set difficulty therefore varies with n_{train} (each size had a different val slice), quietly confounding the historical scaling curve. The confirmatory protocol removes both channels by construction (goal-disjoint splits; Section 3.2).

5.3. Pitfall 3: pseudo-replicated significance

Our original headline significance pooled per-example distances across seeds – treating 3×250 (seed, example) rows as 750 independent paired observations. Predictions of the same example by different seeds are strongly correlated, and examples within an episode share state; the pooled bootstrap therefore overstates evidence. On synthetic nulls matching the data’s dependence structure (episode and example effects shared across seeds; zero true difference), the pooled test rejects 21.2% of the time at nominal 5% (1000 simulations), while the replacement below achieves 5.1%.

Repaired inference. For each contrast we (i) average each example’s paired difference over seeds, (ii) cluster examples by episode, (iii) bootstrap clusters (10,000 resamples) for 95% CIs, and (iv) use an episode-level sign-flip permutation test (10,000 permutations; p -values reported no smaller than Monte Carlo resolution – never $p=0$). **Estimand.** This tests the fixed trained checkpoints on new data from the eval population. It is not calibrated for claims about the training procedure: adding run-level (seed) offsets to the synthetic null drives the episode-clustered test to 55% rejection (and the pooled test to 70%). With only three seeds per variant we therefore release per-seed means for every contrast (in `rerun_analysis.json` and `reanalysis.json`) and make no procedure-level significance claims. The retracted pooled p -values (including a headline $p=0.0002$) should not be cited.

Variant	Mind2Web macro-F1	AITW macro-F1
text_only	0.597 $\pm$ 0.011	0.141 $\pm$ 0.032
vision_text	0.605 $\pm$ 0.016	0.555 $\pm$ 0.036
vision_zeroed	0.469 $\pm$ 0.000	0.110 $\pm$ 0.051
Vision delta	+0.009	+0.414
TF-IDF + LogReg	0.622	0.168
Majority floor	0.472	0.140

Table 2. Stage 1 action-type classification (3 seeds, mean $\pm$ std; exploratory splits). On Mind2Web the instruction text alone contains the action type; on AITW it does not, and vision contributes +0.414 macro-F1. vision_zeroed collapsing to the majority floor is the sanity check that features, not priors, drive the result.

6. Exploratory Results (historical splits, repaired statistics)

All numbers in this section come from the leaky historical splits and are therefore exploratory; they are shown with episode-clustered statistics, and Section 7 gives the prespecified untouched-test versions of every primary claim.

6.1. Stage 1: vision is essential exactly where instructions under-specify the action

Table 2 and Figure 1: on Mind2Web, instructions name the action (“click the submit button”), so vision adds +0.009 macro-F1 over text; on AITW, instructions state goals (“open a new Chrome private window”), text-only collapses toward the majority class, and vision adds +0.414. A TF-IDF ceiling of 0.168 on AITW confirms text is uninformative there regardless of classifier strength.

Unlike the Stage-2 experiments, these Stage-1 runs are not affected by the stream-slicing defect of Section 5.2: they train on each dataset’s own train split and evaluate on its held-out test split (AITW) or test_task split (Mind2Web). We did not, however, audit goal overlap between those official splits, so we treat these numbers as descriptive context rather than as a contamination-checked result. The Stage-1 accuracy that matters for the deployable claim is measured on our own frozen splits in Section 7 (0.795 on the frozen validation split, 0.708 on the untouched test region).

This asymmetry decides where Stage-2 conditioning could matter at all.

6.2. Main ablation under clustered statistics

Table 3 re-scores the historical headline with the repaired statistics (per-seed values in Figure 2). Three observations. First, the auxiliary-loss effect survives:

variant	hit@0.10	L2 $\downarrow$	Δ vs. A [95% CI]	p
A (flat)	0.255 $\pm$ 0.026	0.392	—	—
B (aux loss)	0.305 $\pm$ 0.015	0.362	+0.051 [+0.020, +0.081]	0.0043
C (hard routing)	0.279 $\pm$ 0.039	0.412	+0.024 [-0.016, +0.063]	0.278
D-hook (additive)	0.300 $\pm$ 0.035	0.365	+0.045 [+0.011, +0.085]	0.0227
D-token (prepended)	0.269 $\pm$ 0.031	0.375	+0.015 [-0.018, +0.047]	0.437

Table 3. Exploratory main ablation (AITW all_with_coords, $n_{\text{train}}=1200$, 3 seeds, historical leaky splits). Episode-clustered Δ vs. A with 95% cluster-bootstrap CIs and sign-flip permutation p . Under the repaired analysis, B survives (Holm-adjusted $p = 0.017$ across the four vs.-A contrasts); D-hook is marginal after correction (Holm $p = 0.068$); C and D-token are not significant.

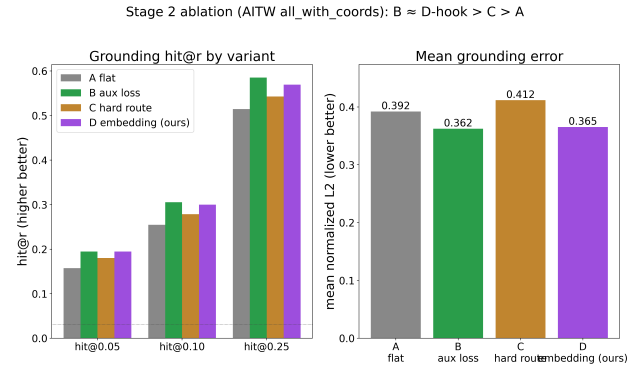


Figure 2. Exploratory ablation on all_with_coords (historical splits; per-seed values). Shown for continuity with the historical record; Table 3 carries the calibrated statistics and Table 5 the confirmatory test-set versions.

B–A on hit@0.10 is +0.051 [+0.020, +0.081] ($p = 0.0043$), where the invalid pooled analysis had claimed $p = 0.0002$ – on this data the conclusion stands, with the evidence overstated by an order of magnitude. (Section 7 shows it does not survive a clean split at all: fixing the statistics was not sufficient.) Second, D-hook weakens to marginal after multiplicity correction. Third, the comparison our proposal cared about most – D-token vs. B – is no longer significant (–0.036 [–0.077, +0.005], $p = 0.114$): on the historical data we cannot even claim the prepended embedding is significantly worse than the auxiliary loss, only that it shows no sign of being better.

6.3. Per-class decomposition: the gain is click disambiguation

Breaking hit@0.10 down by class (Figure 3; deltas below are recomputed on matched seeds 42–44 with parse failures counted as misses, which the original Phase-7 aggregation did neither of): B gains +0.079 on click (68% of examples) at a small scroll cost (–0.014); C gains +0.075 on click but destroys scroll (–0.145) by

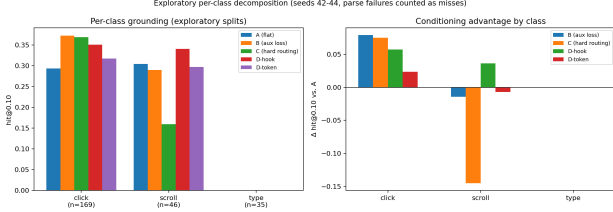


Figure 3. Exploratory hit@0.10 by action class (all_with_coords). The conditioning advantage concentrates in click; type is degenerate on AITW (all variants score exactly 0.000, mean L2 at the $\approx \sqrt{2}$ corner-to-corner distance).

forcing a decode prefix that derails the gesture class; D-hook alone improves both (click +0.057, scroll +0.036). The type class scores exactly 0.000 across all variants with mean normalized L2 at or slightly above $\sqrt{2}$, the corner-to-corner distance (a few predictions land outside the unit square): AITW type events carry degenerate touch coordinates and are ungroundable by any model. This corrects our earlier “type→text-field localization” interpretation: the real signal is click-vs-scroll/type disambiguation for the precise-target class, and the “type” rows merely add a constant drag that cancels in deltas.

6.4. Control conditions

On taps_and_swipes (action type spatially uninformative), the historical 3-seed means show no conditioned variant meaningfully above A (A 0.390 ± 0.008 , D-hook 0.392 ± 0.035 – a difference well inside one seed-std – with B 0.368 ± 0.009 and C 0.325 ± 0.032 at or below A on hit@0.10); per-example logs were not retained for A on this mix, so the clustered version of this control appears only in the confirmatory rerun (Table 6). On Multimodal-Mind2Web (click/type-dominated), D-hook vs. A was -0.006 hit@0.10 / $+0.018$ hit@0.25 over 2 seeds – a null we retain deliberately. On the historical data both of these looked like confirmations that conditioning does not help where action type was assumed to carry no spatial information. Section 7.2 shows that assumption is wrong for the AITW control, which therefore never tested what we believed it tested; the Mind2Web null stands on its own as a cross-dataset result.

6.5. Low-data behavior: no clean curve

Our proposal predicted the advantage is a low-data prior. The historical grid ($n \in \{300, 500, 800\}$, 3 seeds) does not show a clean monotone: B–A on hit@0.10 is +0.067 at $n=300$ but +0.011 at $n=500$ and -0.040 at $n=800$; D-hook is more stable (+0.069, +0.001,

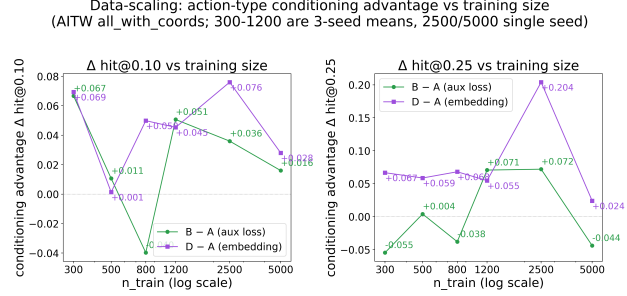


Figure 4. Exploratory conditioning advantage (Δ hit@0.10 vs. A) across n_{train} . Caveat from Section 5.2: each historical size used a different val slice with different goal leakage (up to 85%), so this curve is directional at best; the confirmatory grid (Table 7) fixes one frozen test set for all sizes.

+0.048) and leads hit@0.25 and mean L2 at every low-data size. The two single-seed high-end cells disagree with each other: at $n=5000$ the deltas shrink and partially reverse (B–A hit@0.25 goes negative), while at $n=2500$ D-hook posts the largest conditioning deltas anywhere in the study (+0.076 hit@0.10, +0.204 hit@0.25) – single-seed noise on a differently-leaky val slice, cutting in the direction favorable to conditioning. We report all of this as: inconclusive on the low-data-prior hypothesis – noisy, non-monotone, and confounded by per-size val sets (Figure 4); the prespecified low-data claim is evaluated on the frozen test set in Section 7. An earlier draft of this work called the low-data prediction “confirmed”; that claim did not survive this audit.

6.6. The e2e pipeline claim also weakens

The historical end-to-end result (predicted-type pipeline 0.271 vs. A 0.255 hit@0.10) was previously reported as a win. Under clustered statistics the paired difference is +0.016 [$-0.017, +0.052$] ($p = 0.424$) – not significant; the oracle-type version is marginal (+0.037 [$+0.005, +0.076$]). The deployable-pipeline question is therefore decided by the confirmatory run (Section 7, P4), not the historical data.

6.7. D-token causal-use sensitivity

Feeding the trained D-token model a wrong action id collapses grounding, and zeroing the embedding hurts substantially (Table 4). What this establishes is narrow: the prepended embedding is causally used at inference rather than ignored, so any verdict against it cannot be explained away as an inert conditioning path. Whether it is in fact the worse way to exploit action-type supervision was the course-era conclusion, and Section 7 (P3) reverses it on clean data. Figure 5

contrast	Δ hit@0.10 [95% CI]	Δ hit@0.25 [95% CI]	p
wrong — gold	-0.192 [-0.233, -0.149]	-0.329 [-0.372, -0.284]	$< 10^{-4}$
zero — gold	-0.093 [-0.128, -0.059]	-0.112 [-0.145, -0.079]	$< 10^{-4}$

Table 4. Exploratory D-token sensitivity (episode-clustered, 3 seeds): evaluating the same trained model with wrong or zeroed action embeddings severely degrades grounding. This establishes the embedding is used at inference — a sensitivity/oracle analysis, not evidence that conditioning caused any gain over A.



Figure 5. Image-region attention for a held-out click example under three action-type conditionings (qualitative illustration; screenshot attention mass 0.204 under click vs. 0.132/0.110 under type/scroll).

variant	hit@0.10	L2 ↓	parse	Δ vs. A [95% CI]	p (Holm)
A (flat)	0.219 ± 0.021	0.450	100.0%	—	—
B (aux loss)	0.230 ± 0.005	0.445	100.0%	+0.011 [-0.005, +0.029]	0.23 (0.309)
C (hard routing)	0.219 ± 0.024	0.498	91.7%	+0.000 [-0.022, +0.021]	1
D-hook (additive)	0.276 ± 0.004	0.376	100.0%	+0.057 [+0.035, +0.079]	$< 10^{-4}$ (≤ 0.0004)
D-token (prepended)	0.265 ± 0.004	0.388	100.0%	+0.046 [+0.027, +0.067]	0.0002

Table 5. Confirmatory main ablation on the untouched, goal-disjoint AITW test set ($n=600$ steps; $n_{\text{train}}=1200$; 3 seeds; mean±std across seeds). Δ vs. A: episode-clustered 95% CI and permutation p (Holm-adjusted across the four primary contrasts where applicable).

shows one example where correct conditioning concentrates more attention mass on the screenshot; across the three examples we probed the pattern is not uniform, so we present it strictly as illustration, not evidence.

7. Confirmatory Results (untouched test, pre-specified)

All variants were retrained from scratch on the frozen goal-disjoint splits (identical hyperparameters, seeds 42–44) and evaluated once on the untouched 600-step test set; the four primary contrasts, the headline metric (hit@0.10), the clustered test, and Holm correction were committed before any test evaluation ran (Section 3.2).

P1 (auxiliary supervision) — null. B vs. A on hit@0.10: +0.011 [-0.005, +0.029], permutation $p = 0.23$, Holm $p = 0.309$; mean L2 -0.005 [-0.016, +0.006] ($p = 0.42$). The variant the exploratory analysis identified as the winner does not replicate. Auxiliary action-

variant	hit@0.10	L2 ↓	parse	Δ vs. A [95% CI]	p
A (flat)	0.302 ± 0.009	0.268	100.0%	—	—
B (aux loss)	0.316 ± 0.018	0.264	100.0%	+0.013 [-0.013, +0.041]	0.364
C (hard routing)	0.236 ± 0.064	0.454	84.7%	-0.067 [-0.100, -0.039]	$< 10^{-4}$
D-hook (additive)	0.349 ± 0.020	0.213	100.0%	+0.047 [+0.010, +0.079]	0.0327

Table 6. Confirmatory control mix (taps_and_swipes) on its frozen 400-step test set. The control was designed as the falsifiable half of the thesis — conditioning should not help where action type is uninformative — and it fails: D-hook shows a significant gain here too (Section 7.2).

n_{train}	A	Δ B-A [95% CI]	Δ D-hook-A [95% CI]
$n=300$	0.138	+0.038 [+0.013, +0.064]	+0.049 [+0.025, +0.073]
$n=500$	0.219	-0.004 [-0.027, +0.018]	+0.016 [-0.008, +0.039]
$n=800$	0.256	+0.003 [-0.014, +0.020]	+0.036 [+0.015, +0.057]

Table 7. Confirmatory low-data grid on the same frozen test set for every training size (nested training prefixes), removing the per-size-val-set confound of Figure 4.

type supervision, which adds no information at inference, leaves grounding statistically unchanged on data whose task goals were never seen in training.

P2 (inference-time conditioning) — positive, but oracle. D-hook vs. A: +0.057 [+0.035, +0.079] (Holm $p = \leq 0.0004$; mean L2 -0.074 [-0.100, -0.050]), positive in all three seeds (per-seed deltas +0.035 to +0.085). This is the one large effect that survives the clean split. It is an oracle result: D-hook receives the ground-truth action type at test time, which A never sees. P4 is what happens when that oracle is replaced by a real classifier.

P3 (the proposal’s mechanism) — reversed in our favour, and we flag it as such. D-token vs. B: +0.035 [+0.017, +0.054] (Holm $p = 0.0027$) — the prepended learned action embedding our original proposal advocated significantly outperforms the auxiliary loss on the clean test set. The exploratory data pointed the other way (Section 6), and the course-era version of this work reported the embedding as refuted. On the untouched test both embedding mechanisms beat B (D-token vs. B +0.035 [+0.017, +0.054]; D-hook vs. B +0.046 [+0.026, +0.066], $p = < 10^{-4}$) and both beat A (D-token vs. A +0.046 [+0.027, +0.067], $p = 0.0002$), while hard routing is exactly null (C vs. A +0.000 [-0.022, +0.021], $p = 1$).

Two cautions on reading P3. First, it is an oracle-versus-deployable comparison, not a like-for-like mechanism ranking: D-token and D-hook read the ground-truth action type at inference while B reads nothing, so the embedding variants are being credited with information B never receives. Second, we neither pre-specified nor ran a D-hook versus D-token contrast, so we make no claim that either embedding route is bet-

variant	click ($n=393$)	scroll ($n=136$)	type ($n=71$)
A (flat)	0.312	0.064	0.000
B (aux loss)	0.337	0.042	0.000
C (hard routing)	0.309	0.074	0.000
D-hook (additive)	0.366	0.162	0.000
D-token (prepended)	0.367	0.108	0.000

Table 8. Confirmatory per-class hit@0.10 on the untouched test set (seed-averaged; parse failures counted as misses, so denominators are the fixed class counts in the header). C, D-hook and D-token receive the ground-truth action type at inference; A and B do not. The oracle conditioning gain is proportionally largest on scroll (A 0.064 $\rightarrow$ D-hook 0.162, a $2.5\times$ relative improvement) and modest on click (0.312 $\rightarrow$ 0.366). The auxiliary loss (B) improves click slightly while harming scroll. type remains exactly 0.000 for every variant: AITW type coordinates are degenerate.

ter than the other. What P3 does establish is that the course-era verdict — “the prepended embedding is refuted” — does not survive a clean split. We stress that this reverses in the direction that flatters our own hypothesis, which is precisely why it needed a prespecified test on data we had never touched.

P4 (deployable pipeline) — null. End-to-end predicted-type pipeline vs. A: +0.015 [-0.004, +0.034] (Holm $p = 0.309$). The reason is Stage 1 generalization, not Stage 2: the action-type classifier scores 0.795 accuracy on the frozen validation split but only 0.708 on the untouched test region, and the paired predicted-vs-oracle gap is -0.048 [-0.061, -0.035] ($p = <10^{-4}$). That gap absorbs 76% of the oracle arm’s advantage over A (0.282 ± 0.007 versus 0.219 ± 0.021 , i.e. +0.063), leaving a residual that is statistically indistinguishable from zero. Stage-1 error leaves no detectable benefit, directly contradicting the historical claim that “classifier error is not the bottleneck.” We note what this drop is not: both the frozen validation split and the untouched test region are goal-disjoint from training, so novel task goals alone cannot explain the difference between them. The two held-out sets differ in where they sit in the episode stream (the test region lies entirely beyond index 5,500), and we did not run the ablations needed to isolate that cause, so we report the drop as an observed generalization gap rather than a diagnosed one. Either way it is the binding constraint on the deployable pipeline.

7.1. Where the confirmatory gain lives: proportionally scroll, absolutely still click

Table 8 revises the exploratory mechanism story of Section 6.3. On leaky data the conditioning advantage looked like click disambiguation. On the untouched test set the largest relative effect is on scroll (+0.098

absolute, $2.5\times$ relative) versus a smaller proportional click gain (+0.053, $1.2\times$). Because click is roughly three times as frequent in the test set, click nevertheless still supplies 61% of the aggregate gain against scroll’s 39%: what moved is where the effect is concentrated, not which class contributes the most additional hits. These per-class figures are seed-averaged point estimates without interval estimates — the prespecified analysis covers the aggregate contrasts only — so we read them as a descriptive decomposition, not as a tested mechanism claim. The mechanism this supports is different and simpler: swipe targets occupy a narrow, largely goal-independent region of the screen, so being told “this step is a scroll” relocates the prediction in a way the flat baseline cannot infer from the screenshot and instruction alone. B, which gets the same supervision only as a training signal, captures none of it and slightly harms scroll.

7.2. The control condition does not behave as designed

Our prespecified control assumed that on taps_and_swipes action type carries no spatial information, so conditioning should not help. It does help: D-hook vs. A on the control test set is +0.047 [+0.010, +0.079] ($p = 0.0327$), while B is null (+0.013 [-0.013, +0.041], $p = 0.364$) and C is significantly harmful (-0.067 [-0.100, -0.039], $p = <10^{-4}$). These control contrasts were prespecified as secondary, and their p -values are unadjusted; we use them to withdraw a scope claim rather than to establish one, which is the direction in which an unadjusted secondary result can safely be read. We report this as a failed control, not a second confirmation. It is coherent with Section 7.1: if the exploitable signal is “swipes land in a predictable region,” then a tap-versus-swipe mix is exactly a setting where action type is spatially informative, and our control was mis-designed rather than uninformative. The consequence for the paper is concrete: we cannot claim “conditioning helps if and only if action type is spatially predictive,” because we no longer have a matched condition in which it demonstrably is not.

7.3. Low data

The confirmatory low-data grid (Table 7) evaluates every training size against the same frozen test set, removing the per-size-val-set confound that made Figure 4 unreadable. D-hook’s oracle advantage is present at $n=300$ and $n=800$ and not resolved at $n=500$; B is positive only at $n=300$. There is no clean monotone low-data trend in either variant, so we again decline the “low-data prior” claim our proposal predicted.

Secondary contrasts, per-class decompositions, parse rates, and per-seed values for every run are in the released results/phase9_rerun/rerun_analysis.json, which also carries the per-seed mean for each side of every contrast. No procedure-level significance claim is made from three seeds (Section 5.3).

7.4. What the broken evaluation actually cost

Because the same five-variant ablation was run under both protocols, we can separate the two defects and ask what each one cost. Three states are distinguishable: (i) the original analysis (leaky split, pooled bootstrap) — what a course-era version of this paper reported; (ii) the statistically repaired analysis on the same leaky data (Section 6); and (iii) the confirmatory analysis on the untouched split.

- Which variant wins. (i) B beats A, $p=0.0002$. (ii) B still beats A, $+0.051$ [$+0.020$, $+0.081$], $p=0.0043$. (iii) B is null ($+0.011$ [-0.005 , $+0.029$]); the embedding mechanisms are the only ones that help.
- Whether our own hypothesis was refuted. (i) D-token is significantly worse than B (pooled $p=0.0192$), so the proposal’s prepended embedding is refuted. (ii) the gap is no longer significant (-0.036 [-0.077 , $+0.005$], $p=0.114$) — the refutation was already an artifact of the pooled test. (iii) D-token significantly beats B ($+0.035$ [$+0.017$, $+0.054$]).
- Where the gain lives. (i)–(ii) click disambiguation. (iii) the relative effect is largest on scroll ($2.5\times$ versus $1.2\times$ for click), though click’s higher frequency still makes it 61% of the absolute gain.
- Whether the pipeline is deployable. (i) the predicted-type pipeline was reported as beating A, with “classifier error is not the bottleneck” — a descriptive comparison that no test ever supported. (ii) any calibrated test shows it was never significant ($+0.016$ [-0.017 , $+0.052$], $p = 0.424$). (iii) the pipeline is null, with Stage-1 error absorbing 76% of the oracle gain.
- Whether the control holds. (i)–(ii) conditioning looks neutral where action type was assumed uninformative. (iii) the control shows a significant effect, so it is not a valid control.

Two lessons separate cleanly. First, the statistical repair alone was not sufficient: it withdrew two unsupported claims (the D-token refutation and the deployable pipeline) but left the headline “auxiliary supervision works” and the click-mechanism story intact

and wrong — those required the clean split. Fixing inference without fixing the data would have produced a paper that was better calibrated and still substantively mistaken. Second, the direction of error is not systematic: the flawed protocol inflated some effects, suppressed our own hypothesis, and mis-located the mechanism. A study can be internally consistent, multi-seed, ablated, controlled, and significant at the pooled $p=0.0002$ we now retract, and still have every one of its mechanistic claims move under a clean split and a calibrated test.

8. Limitations

Every positive effect among the four prespecified contrasts is oracle-gated: D-hook and D-token receive ground-truth action types that a deployed agent would have to predict, and when we predict them the effect disappears (P4). No headline claim in this paper should be read as a deployable improvement. One non-oracle positive does appear among the secondary results — B over A at $n_{\text{train}}=300$ (Table 7) — but it is a secondary, unadjusted contrast in a grid where the same comparison is null at both larger sizes, so we do not build a claim on it. Our control condition failed (Section 7.2), so we have no valid matched setting in which action type is demonstrably uninformative; the scope condition on when conditioning helps is therefore untested, and designing a genuine control (for instance, targets randomized within class) is future work. The scroll-versus-click mechanism reading in Section 7.1 is a post-hoc decomposition of confirmatory data, not a prespecified test.

Offline step-level hit@radius is not long-horizon task success: it scores logged single steps, without interaction, recovery, or compounding-error dynamics — the motivating failure mode is measured here only via its step-level signature. The backbone is a single 2B model; conclusions are about this scale and training recipe, not about GUI agents generally, and our baselines are small matched-compute models trained by us, not state-of-the-art systems. The confirmatory estimand is fixed checkpoints, not the training procedure (three seeds; per-seed effects reported). AITW’s type coordinates are degenerate, so “grounding” on all_with_coords is effectively click/scroll grounding plus a constant; the click-concentration analysis is post-hoc on exploratory data and only its prespecified aggregate versions are confirmatory. Goal strings are our dependence proxy; 17 near-duplicate train/test goal pairs (disclosed in the manifest) survive normalization, and subtler sharing (same app, similar screens) is not modeled. The Mind2Web control used 2 seeds without per-example logs. Finally, the mirror we pin

is a third-party repackaging of AITW; we verified revision stability but not byte-level provenance against the original release.

9. Conclusion

On a test set whose task goals were never seen in training, the factorization thesis survives only in oracle form. Telling a 2B GUI agent the ground-truth action type improves grounding (+0.057 [+0.035, +0.079] hit@0.10). Both mechanisms that consume that oracle signal beat the auxiliary loss, including the prepended action embedding our proposal advocated (D-token+B +0.035 [+0.017, +0.054]) — a reversal of the course-era verdict that the embedding was refuted, though the comparison is oracle-versus-deployable, and we neither prespecified nor ran a D-hook versus D-token contrast, so we do not claim either embedding mechanism is better than the other. But auxiliary supervision alone is null, hard routing is null, the largest relative effect moves to scroll rather than the click disambiguation we previously reported (click still supplies 61% of the absolute gain), our control condition turns out not to be a control, and the only configuration one could actually deploy — predicting the action type and conditioning on the prediction — is statistically indistinguishable from doing nothing, because the classifier drops from 0.795 to 0.708 accuracy between the frozen validation split and the untouched test region. The honest summary is that we measured a real but oracle-gated effect and no deployable one.

The methodological result is the durable one. We ran the same five-variant ablation twice: once under a protocol with three defects (an `inputs_embeds/M-RoPE` interaction fabricating an 8σ false negative, goal-level split leakage reaching 85%, and a pooled bootstrap with 21.2% measured type-I error), and once under a repaired one. The defects did not merely exaggerate significance; they inverted which mechanism wins, which class carries the gain, whether our hypothesis was refuted, whether the pipeline was deployable, and whether the control held. None of the three is exotic — stream-sliced splits, embedding-injection shortcuts, and pooled per-seed bootstraps are ordinary practice — so we expect the same reversals are latent in other offline GUI-grounding evaluations. The frozen splits, per-example predictions, clustered-inference code, and prespecification are released together with the paper so that both the object-level result and the audit can be checked and rerun.

Contributions

A. Chauhan: training and evaluation infrastructure, Stage-2 conditioning variants, cloud experiment orchestration, statistical reanalysis and confirmatory protocol, writing. A. Ilyasov: Stage-1 classifier study, data pipeline and error analysis, figures, editing. Both authors contributed to problem formulation and experimental design and approved the final manuscript.

Generative AI Usage Statement

Large-language-model assistants (Anthropic Claude) were used extensively under the authors’ direction: writing and debugging training, evaluation, and statistical-analysis code; orchestrating cloud experiments; drafting the integrity audit that led to Sections 5 and 7; and drafting and editing this manuscript. The authors specified the research questions and methodology, reviewed the analyses, verified reported numbers against the committed result artifacts (every confirmatory and reanalysis table in this paper is generated programmatically from those artifacts by `scripts/p10_render_paper_tables.py`; the Stage-1 summary table is hand-entered from the Phase-2/3 result files), and take full responsibility for all content.

References

- [1] M. Ahn et al. Do as I can, not as I say: Grounding language in robotic affordances. In CoRL, 2022.
- [2] A. Brohan et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. arXiv preprint arXiv:2307.15818, 2023.
- [3] K. Cheng et al. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. arXiv preprint arXiv:2401.10935, 2024.
- [4] X. Deng et al. Mind2Web: Towards a generalist agent for the web. In NeurIPS, 2023.
- [5] B. Gou et al. Navigating the digital world as humans do: Universal visual grounding for GUI agents. arXiv preprint arXiv:2406.01729, 2024.
- [6] K. Q. Lin et al. ShowUI: One vision-language-action model for GUI visual agent. arXiv preprint arXiv:2411.17465, 2024.
- [7] Z. Lu et al. UI-R1: Enhancing action prediction of GUI agents by reinforcement learning. arXiv preprint arXiv:2503.21620, 2025.
- [8] T. Luo, L. Logeswaran, J. Johnson, and H. Lee. Visual test-time scaling for GUI agent grounding. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025. arXiv:2505.00684.
- [9] C. Rawles et al. Android in the wild: A large-scale dataset for android device control. In NeurIPS, 2023.

- [10] P. Wang et al. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. arXiv preprint arXiv:2409.12191, 2024.
- [11] Z. Wu et al. OS-Atlas: A foundation action model for generalist GUI agents. arXiv preprint arXiv:2410.23218, 2024.
- [12] S. Yao et al. ReAct: Synergizing reasoning and acting in language models. arXiv preprint arXiv:2210.03629, 2022.
- [13] Y. Yao et al. UI-TARS: Pioneering automated GUI interaction with native agents. arXiv preprint arXiv:2501.12326, 2025.
- [14] Q. Zheng et al. GUI-Actor: Coordinate-free visual grounding for GUI agents. arXiv preprint arXiv:2503.15197, 2025.